

FINAL EXPLANATION: PHYSICS GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: PHYSICS GRADE 10: 3.3 INTRODUCTION TO ELECTRONICS

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation. Use your own words to answer the Driving Question: Why do some materials conduct electricity while others do not, and how does this affect the safety and efficiency of household wiring and of devices such as solar panels, computers and MRI machines? Use the evidence from your five-material bulb test, the resistivity data, your class model, the Driving Question Board (DQB) and the Kenyan case studies. Write your answers in the space under each prompt. Use the words conductor, insulator, semiconductor and superconductor carefully, and give numbers and units where you can. Show your thinking in full sentences. Use the data table in Section 2 when it is asked for.

Section 1: What the five-material test showed

Mr Otieno, an electrician in Nairobi, wires a house with copper cable covered in PVC plastic and can safely hold the insulated wire while it carries current at 240 V. In class you tested copper wire, glass, dry wood, plastic and a silicon diode (1N4007) in a 3 V circuit with a torch bulb, always in that order.

(a) Describe what happened to the bulb for each of the five materials, including both ways of connecting the diode.

(b) How many of the four ordinary materials (copper, glass, wood, plastic) let the bulb light? What was different about the diode?

(c) Use your results to explain why Mr Otieno can safely hold the plastic-covered copper cable.

(a) Copper wire: the bulb was bright. Glass: the bulb was off. Dry wood: the bulb was off. Plastic strip: the bulb was off. Silicon diode (1N4007): the bulb lit when it was forward-connected (forward-biased) and was off when it was reverse-connected (reverse-biased).

(b) Only 1 of the 4 ordinary materials, copper, let the bulb light. The diode was different from all of them because it conducts in one direction only.

(c) Copper is a conductor, so it carries current easily along the wire. Plastic (PVC) is an insulator that lets almost no current through, so the current stays inside the copper and almost none passes to Mr Otieno's hand. This makes the plastic cover a safe layer for the cable even at 240 V. Insulators allow almost no current, not exactly none.

Section 2: Using resistivity data to classify materials

The table shows approximate resistivity values at about 20 °C.

Material	Resistivity ρ (ohm metre)
Copper	1.7×10^{-8}
Germanium (pure)	0.46
Silicon (pure)	2.3×10^3
Dry wood	about 10^8 to 10^{11}
Glass	about 10^{10} to 10^{14}
PVC	about 10^{13} or more

- (a) Put the materials in order from best to worst conductor.
(b) Sort them into conductor, semiconductor and insulator.
(c) Calculate how many times more resistive pure silicon is than copper. Show your working.
(d) How is conductivity related to resistivity?
(e) Use the data to explain why household wiring uses copper covered in PVC.

- (a) Best to worst conductor (lowest to highest resistivity): copper, germanium, silicon, dry wood, then glass and PVC (the highest).
(b) Conductor: copper. Semiconductors: germanium and silicon. Insulators: dry wood, glass and PVC.
(c) Silicon / copper = $(2.3 \times 10^3) / (1.7 \times 10^{-8}) = \text{about } 1.4 \times 10^{11}$. Silicon is about 1.4×10^{11} times (about 10^{11} times) more resistive than copper.
(d) Conductivity is the reciprocal of resistivity: conductivity = $1/\rho$.
(e) Copper has a very low resistivity (1.7×10^{-8} ohm metre), so it carries current efficiently with little energy wasted. PVC has a very high resistivity (about 10^{13} ohm metre or more), so it allows almost no current to leave the wire. Together they make wiring efficient and safe for people to handle.

Section 3: Why materials differ: energy gaps, temperature and doping

All the materials you tested are made of atoms with electrons, yet they behave very differently. (a) Using the idea of energy bands and the energy gap (Eg), explain why copper conducts but glass, wood and plastic almost do not. Give the gap values you used. (b) Why does pure (intrinsic) silicon conduct a little, and what are the holes that appear? How does warming change its conductivity? How is this different from copper? (c) Explain how doping silicon with phosphorus and with boron changes its conduction. Name the type of semiconductor and the majority carriers in each case. Is doped silicon charged? (d) Why is doping useful, and how does a p-n junction explain the result you saw with the diode and the bulb?	(a) In copper (a metal) the bands overlap, so the energy gap is 0 eV and outer electrons are free to move through the metal as free electrons. In glass, wood and plastic the energy gap is large (above about 5 eV), so almost no electrons gain enough energy to move, and almost no current flows. (b) Pure silicon has a small gap (1.1 eV; germanium 0.67 eV). At room temperature a few electrons gain enough energy to cross the gap and become free electrons. Each leaves behind a hole (a missing electron), making an electron-hole pair. Warming makes more pairs, so the conductivity of silicon rises. In copper, the resistance rises slightly when warmed. (c) Phosphorus has 5 valence electrons, so it adds an extra free electron: n-type, majority carriers electrons. Boron has 3 valence electrons, so it leaves a hole: p-type, majority carriers holes. Doped silicon is still electrically neutral overall. (d) Typical doping is about 1 atom in 10^7 to 10^8, which raises conductivity by a factor of many thousands, so conduction can be controlled. Joining p-type and n-type gives a p-n junction (a diode), which conducts when forward-biased (bulb lit) and blocks current when reverse-biased (bulb off).
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Section 4: Superconductors and MRI machines

Ms Achieng showed a YBCO disc cooled in liquid nitrogen at 77 K (-196 C) making a magnet float. The critical temperatures (Tc) are: mercury 4.2 K, NbTi 9.2 K, YBCO about 92 K. (a) What is special about the resistance of a superconductor below its Tc, and how is this different from copper? (b) Dr Wekesa operates an MRI machine at Kenyatta National Hospital. Which coolant (liquid nitrogen at 77 K or liquid helium at 4.2 K) is needed for the NbTi magnet? Explain using the	(a) Below its critical temperature (Tc) a superconductor has zero resistance. Copper always has some resistance (1.7×10^{-8} ohm metre at about 20 C), so some energy always becomes heat. (b) Liquid helium at 4.2 K is needed. NbTi has $T_c = 9.2$ K, and 4.2 K is below 9.2 K, but liquid nitrogen at 77 K is far above 9.2 K, so NbTi would not be superconducting. (YBCO, with T_c about 92 K, works at 77 K.) (c) With zero resistance, the NbTi coils carry a very large current without energy lost as heat, giving the strong magnetic field of about 1.5 T to 3 T. Household wiring stays copper and PVC because room temperature is far above every T_c we studied.
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data. (c) Why does an MRI magnet need a superconductor, and why is household wiring still copper and PVC?	
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Section 5: Final answer to the Driving Question	
Write a clear explanation that answers the Driving Question: Why do some materials conduct electricity while others do not, and how does this affect the safety and efficiency of household wiring and of devices such as solar panels, computers and MRI machines? Your answer must: – use at least two pieces of evidence from your tests or data (include numbers and units); – use the words conductor, insulator, semiconductor and superconductor correctly; – link to Mr Otieno's wiring and to at least two of these Kenyan cases: the Garissa solar plant (50 MW), microchips in phones, M-Pesa and computers, and the MRI at Kenyatta National Hospital; – say how your class model changed from Lesson 1 to now.	Materials differ in whether they let current flow because of their energy bands. In copper the bands overlap (gap 0 eV), so its outer electrons are free to move, and its resistivity is very low (1.7×10^{-8} ohm metre). In glass, wood and plastic the gap is above about 5 eV, so almost no electrons can move and resistivity is very high (PVC about 10^{13} ohm metre or more). In our bulb test, only copper out of the four ordinary materials lit the bulb. This is why Mr Otieno's Nairobi wiring uses copper for efficient current flow and PVC as an insulator, so he can safely hold the cable at 240 V. Silicon and germanium are semiconductors with small gaps (1.1 eV and 0.67 eV), so a few free electrons and holes form and they conduct a little, more when warmer. Doping with phosphorus (n-type) or boron (p-type), about 1 atom in 10^7 to 10^8, raises conductivity thousands of times in a controlled way. A p-n junction conducts one way only (our diode lit the bulb forward-biased and not reverse-biased). This is the basis of silicon solar panels such as the 50 MW Garissa solar plant and of the microchips with billions of transistors in phones, M-Pesa and computers. Superconductors have zero resistance below their critical temperature. At Kenyatta National Hospital, the MRI uses NbTi coils (T_c 9.2 K) cooled to 4.2 K by liquid helium, so large current flows without energy loss and gives a field of about 1.5 T to 3 T. My class model changed from only showing electrons and atoms in Lesson 1 to showing energy bands and gaps, free electrons, holes, doping and critical temperature, which explains the differences.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence and data	Uses accurate results from the bulb test and resistivity, gap and T_c values with	Uses most evidence and numbers correctly, with a few missing units or	Uses little evidence, or numbers and

	correct units and the correct order of materials.	small slips.	results are missing or incorrect.
Scientific explanation using the band model	Clearly explains conductors, insulators, semiconductors, temperature effects, doping and p-n junctions using energy gaps, free electrons and holes.	Explains most ideas using gaps and free electrons or holes, with some gaps in reasoning.	Describes what happens but does not explain why, or confuses ideas such as n-type and p-type.
Correct use of terms and conventions	Uses conductor, insulator, semiconductor, superconductor, forward-biased and reverse-biased correctly and says insulators allow almost no current.	Mostly correct terms, with one or two mix-ups.	Often mixes terms or claims insulators allow no current at all.
Connection to Kenyan contexts and the Driving Question	Links the answer to Mr Otieno's wiring and at least two Kenyan cases and answers safety and efficiency fully.	Links to wiring and one other case, answering the Driving Question in part.	Few or no links to the contexts, or the Driving Question is not answered.
Model revision and clarity	Clearly describes how the class model changed from Lesson 1 and writes in well-organised sentences.	Mentions model change and writes clearly most of the time.	Does not describe model change, or writing is hard to follow.

ATTRIBUTION AND LICENSING

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